Endogenous Reachability Collapse (ERC)

A Minimal Dynamical Model for Loss of Recoverability Without Loss of State Space

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Abstract

We present a minimal dynamical system in which loss of recoverability emerges endogenously without removal of the underlying state space.

The model consists of a slow–fast system in which an internal constraint variable dynamically modulates the effective basin of attraction of a base state. Numerical simulations suggest that the fraction of initial conditions that return to the base attractor decreases as the internal constraint increases, while the attractor itself remains present.

We further demonstrate that trajectories can self-induce loss of recoverability by driving the accumulation of internal constraint, leading to a dynamic contraction of the reachable set. In addition, the system exhibits path-dependent behavior consistent with hysteresis-like dynamics, where recovery is not restored even after constraint reduction.

A state-dependent recoverability threshold is identified, separating initial conditions that return from those that do not. Within the explored parameter range, this threshold decreases approximately linearly with initial constraint.

These results are consistent with the hypothesis that loss of recoverability may arise from reduced accessibility rather than from elimination of states. The model is intentionally minimal and is presented as a falsifiable proof-of-principle.

1. Introduction

In many dynamical systems, failure is typically interpreted as the disappearance of stable states or attractors. However, systems may instead retain their underlying state space while losing the ability to access certain regions of it.

This work explores the hypothesis that:

Loss of recoverability may arise from loss of reachability, even when the attractor remains present.

Such a mechanism may be relevant in systems where internal variables accumulate over time, modifying accessibility without requiring structural changes to the system itself.

To investigate this, we construct a minimal dynamical model in which accessibility emerges from the interaction between system trajectories and an internal constraint variable.

2. Model Definition

We consider a system with:

- Two fast variables: x(t), y(t)
- One slow internal variable: c(t)

Define:

$$r^2 = x^2 + y^2$$

$$\dot{x} = g(r,c)\,x - \omega y$$
$$\dot{y} = g(r,c)\,y + \omega x$$

with:

$$g(r,c) = -(r^2 - a(c)^2)(r^2 - b^2)$$

The recoverability boundary is defined as:

$$a(c) = a_0 - \alpha c$$

where a_0 defines the baseline recoverability radius and α controls the strength of constraint-induced contraction.

The internal constraint evolves as:

$$\dot{c} = \varepsilon\left(\frac{r^2}{1 + r^2} - c\right)$$

3. Interpretation

The system contains:

- A base attractor at r = 0
- An outer attractor at r = b
- A moving recoverability boundary a(c)

For fixed c:

- r < a(c) → trajectories return to the base attractor
- r > a(c) → trajectories converge to the outer attractor

Crucially:

- The attractor persists across all conditions
- Accessibility depends on system history through c(t)

4. Results

4.1 Basin contraction

Numerical simulations suggest that the basin of attraction of the base state contracts as internal constraint increases.

The fraction of initial conditions that return decreases with c, while the attractor itself remains present.

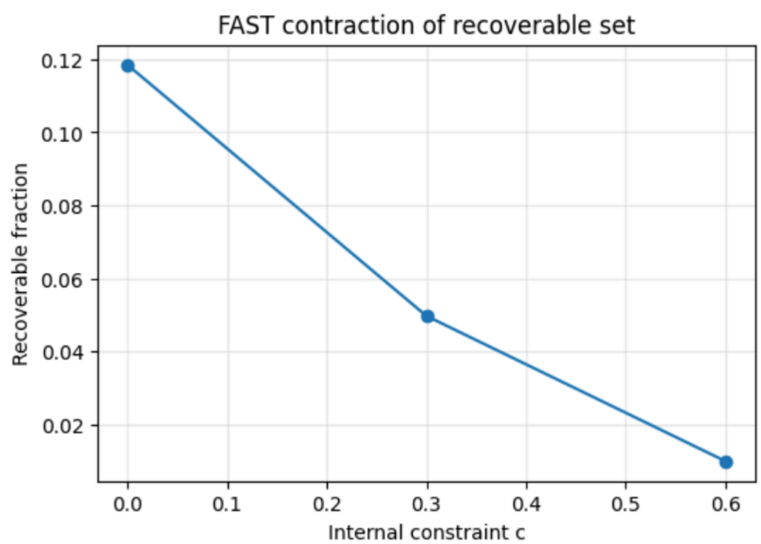

Figure 1. *Contraction of the recoverable set under increasing internal constraint.*

The fraction of initial conditions that return to the base attractor is shown as a function of the internal constraint c. The recoverable fraction decreases monotonically as c increases, suggesting a progressive reduction in dynamic accessibility while the attractor itself remains present.

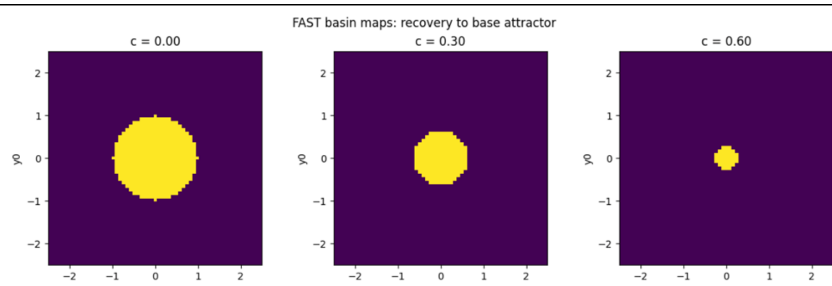

Figure 2. *Basin of attraction of the base state for increasing internal constraint.*

Each panel shows the set of initial conditions that return to the base attractor (yellow) for representative values of the internal constraint c. The basin progressively contracts as c increases, indicating loss of accessibility without removal of the attractor.

4.2 Quantitative characterization

The contraction of the recoverable set can be quantified by the fraction of initial conditions that return, providing an operational measure of accessibility.

4.3 State-dependent recoverability threshold

We identify a critical initial radius r_0^* separating recoverable and non-recoverable trajectories.

The critical radius r_0^* is defined operationally as the largest initial radius for which trajectories return to the base attractor within the simulation time window.

This threshold depends on the initial internal constraint c_0.

Numerical results suggest that:

- Higher c_0:→ lower r_0^*
- Increased vulnerability under pre-existing constraint

Within the explored range:

$$r_0^* \approx a_0 - k\, c_0$$

where k is an empirically estimated proportionality constant within the explored parameter range.

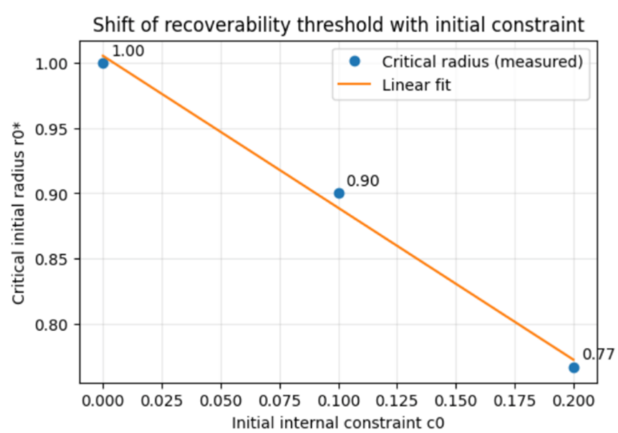

***Figure 3**. State-dependent recoverability threshold as a function of initial constraint.*

The critical initial radius r_0^* separating recoverable and non-recoverable trajectories is plotted against the initial internal constraint c_0. Within the explored range, the threshold appears to decrease approximately linearly, within the explored parameter range, indicating increased vulnerability under pre-existing constraint.

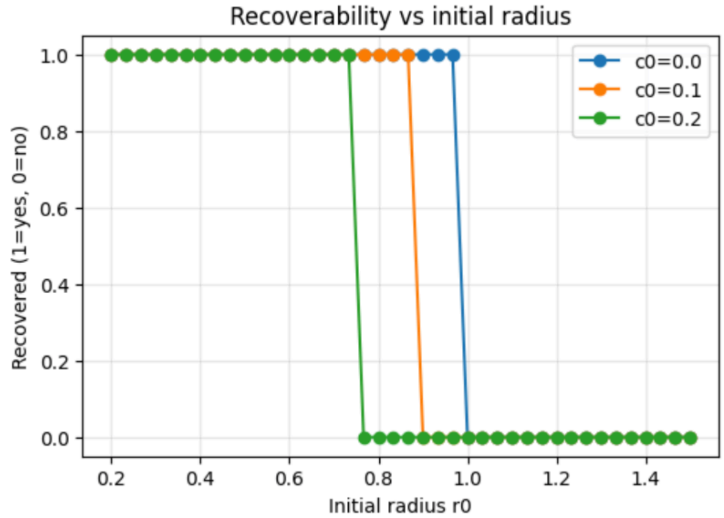

Figure 4. *Recoverability as a function of initial radius for different constraint levels.*

Binary recovery outcomes (1 = recovery, 0 = no recovery) are shown as a function of the initial radius r_0 for different values of initial constraint c_0. The transition from recovery to non-recovery sharpens as constraint increases, consistent with a state-dependent boundary.

4.4 Path-dependent loss of recoverability

Simulations show that trajectories can drive their own loss of recoverability by increasing internal constraint.

Even after constraint is reduced:

- The recoverability boundary partially re-expands
- But trajectories may not return

This behavior is consistent with hysteresis-like dynamics.

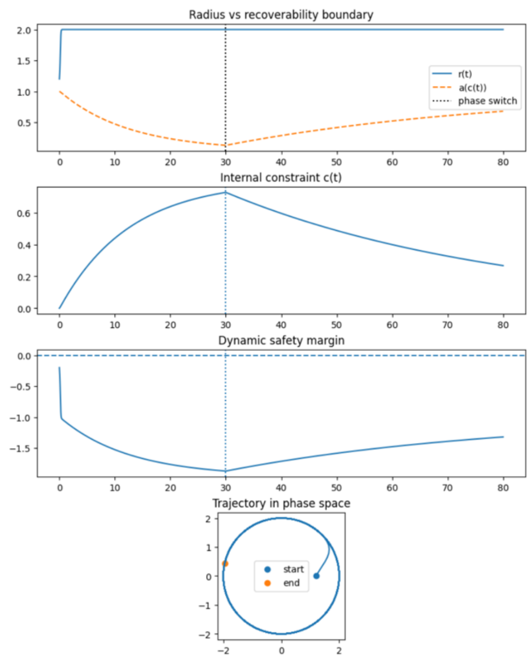

Figure 5. *Path-dependent loss of recoverability under endogenous constraint dynamics.*

Top to bottom: radius r(t) and moving recoverability boundary a(c(t)), internal constraint c(t), dynamic safety margin a(c(t)) − r(t), and trajectory in phase space. The system is initially driven into a high-constraint regime, leading to contraction of the recoverability boundary. Even after constraint is reduced, the trajectory does not return to the base attractor, consistent with hysteresis-like behavior and path-dependent loss of accessibility.

The vertical dashed line indicates the time at which constraint accumulation is reduced.

5. Key Observation

The results are consistent with the interpretation that systems may lose access to an attractor without removal of the attractor itself.

6. Limitations

This model is intentionally minimal and should be interpreted as a proof-of-principle.

- Functional forms are phenomenological
- No direct mapping to a specific biological system is claimed
- Irreversibility is observed operationally, not proven mathematically
- Linear relationships are approximate and limited to explored ranges

7. Methods

Numerical simulations were performed using standard ordinary differential equation (ODE) integration methods with fixed time stepping.

The dynamical system is defined as described in Section 2, with a slow–fast structure coupling radial dynamics and an internal constraint variable.

The following parameter values were used in the simulations:

- $\alpha_0 = 1.0$
- $\alpha = 1.0$
- $b = 2.0$
- $\epsilon = 0.05$
- $\omega = 1.0$

These values were chosen to produce a clear separation between recoverable and non-recoverable regimes while maintaining numerical stability. No claim is made that these values correspond to a specific physical system.

Initial conditions were sampled over a uniform grid in the (x_0, y_0) plane.

For basin mapping, the initial internal constraint was fixed at selected values c_0, and the system was simulated forward in time for each initial condition.

Recovery criterion

A trajectory was classified as recovered if it returned to a neighborhood of the base attractor (defined operationally as r(t) remaining within a small neighborhood of zero (i.e., $r(t) < \delta$)) within a fixed simulation time window.

Conversely, trajectories that did not return within this window were classified as non-recovering.

This operational definition does not imply strict mathematical irreversibility, but provides a consistent numerical criterion for distinguishing regimes.

Basin estimation

The recoverable set was estimated as the fraction of sampled initial conditions that satisfied the recovery criterion.

This allowed construction of:

- Basin maps across (x_0, y_0)
- Recoverable fraction as a function of internal constraint c
- State-dependent thresholds r_0^*

Reproducibility

All simulations are deterministic and fully reproducible from the provided code and parameters.

The implementation is intentionally minimal and designed to enable independent replication and falsification of the reported results.

8. Discussion

The mechanism described here may be relevant in classes of systems where:

- Internal constraints accumulate
- Recovery depends on system history
- Accessibility is dynamically modulated

Potential domains include:

- Metabolic systems
- Redox dynamics
- Neural systems
- Complex adaptive systems

Further work is required to establish empirical grounding.

9. Conclusion

We have presented a minimal dynamical system in which:

- The state space remains intact
- A base attractor persists
- But access to that attractor collapses dynamically

These results are consistent with the working hypothesis that:

Loss of recoverability may emerge from loss of reachability rather than loss of state space.